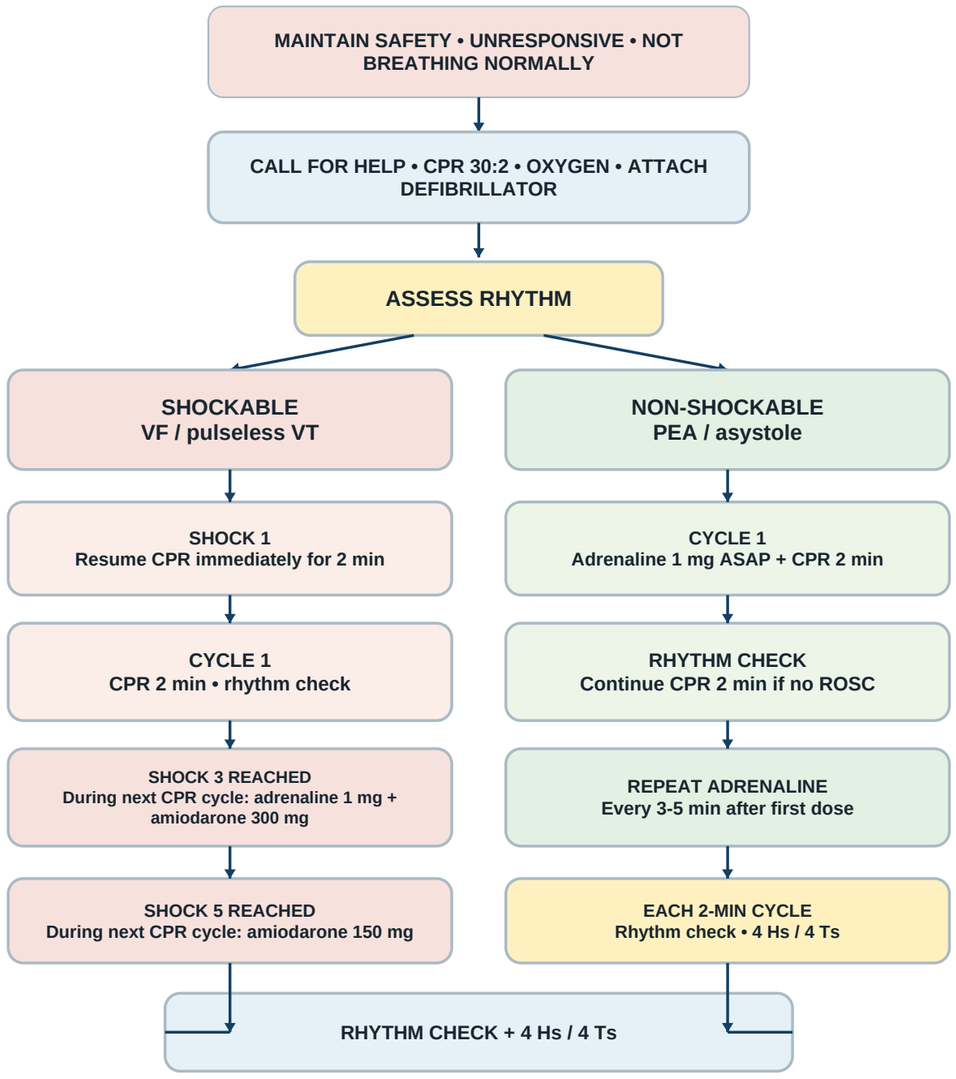
ADULT ADVANCED LIFE SUPPORT - ONE-PAGE COGNITIVE AID

UK paramedic practice | Core 2-minute cycle, exact arrest medicines, reversible causes, special circumstances and ROSC

1. CORE ALS ALGORITHM - CONTINUE 2-MINUTE CYCLES



HOW THE CYCLES + DRUG TIMING WORK

[ ] Shockable VF/pVT: shock, then CPR for 2 min. Count the number of shocks, not rhythm checks.
[ ] After shock 3, give adrenaline 1 mg + amiodarone 300 mg during the following CPR cycle.
[ ] After shock 5, give amiodarone 150 mg during the following CPR cycle. Continue adrenaline every 3-5 min.
[ ] Non-shockable PEA/asystole: give adrenaline 1 mg as soon as IV/IO access is available, then every 3-5 min.
[ ] At every 2-min rhythm check: if VF/pVT, shock; if PEA/asystole, resume CPR without shock; if ROSC, move to post-ROSC care.
[ ] CPR 100-120/min, depth 5-6 cm, full recoil, minimal pauses. Rotate compressor.
[ ] Waveform capnography. If ETCO2 <1.5 kPa, ask why: CPR quality, ventilation, airway/circuit or reversible cause?

2. REVERSIBLE CAUSES - TICK + TREAT

[ ] Hypoxia: open airway; oxygenate and ventilate.
[ ] Hypovolaemia: control haemorrhage/loss; replace volume.
[ ] Hyper/hypokalaemia + metabolic: test and treat.
[ ] Hypo/hyperthermia: measure core temperature; correct.
[ ] Thrombosis: coronary or pulmonary pathway.
[ ] Tension pneumothorax: immediate decompression.
[ ] Toxins: antidote/supportive care; seek expert advice.
[ ] Tamponade: trauma/specialist pathway.

3. SPECIAL CIRCUMSTANCES - WHAT CHANGES?

WITNESSED, MONITORED VF/pVT: STACKED SHOCKS
- Only when VF/pVT arrest or re-arrest is witnessed and monitored by the crew, with a defibrillator immediately available.
- Up to 3 stacked shocks may be considered. Check rhythm after each shock; stop if ROSC or rhythm changes.
- For adrenaline/amiodarone timing, the whole initial stacked sequence counts as shock 1, not shocks 1-3.

HYPOTHERMIA: KEEP 2-MIN CYCLES
- <30°C: continue CPR/rhythm checks; maximum 3 shocks, then defer further shocks until >30°C.
- <30°C: withhold adrenaline. If shockable, one amiodarone 300 mg loading dose may be given; defer further doses.
- 30-35°C: resume shocks; adrenaline every 6-10 min, not every alternate cycle.
- >35°C: standard 3-5 min drug timing. Prevent heat loss, rewarm and seek ECPR early.
- Target normocapnia; hypothermia may require lower minute ventilation.

TORSADES DE POINTES
- Pulse: magnesium sulfate 2 g IV over 10 min; correct K+/Mg2+; seek expert help.
- Pulseless: treat as VF/pVT; defibrillate and give magnesium 2 g IV/IO.

PREGNANCY >20 WEEKS
- Manual left uterine displacement; standard shocks and ALS drugs.
- Early obstetric/surgical/neonatal help; prepare resuscitative hysterotomy as soon as possible.

DROWNING / TRAUMA / ANAPHYLAXIS
- Drowning: 5 initial breaths with 100% oxygen; apply hypothermia changes if cold.
- Trauma: treat haemorrhage, hypoxia, bilateral tension pneumothorax and tamponade in parallel.
- Anaphylaxis before arrest: IM adrenaline + IV crystalloid; in arrest use standard ALS adrenaline.

TOXINS: NALOXONE
- Opioid respiratory depression WITH a pulse: ventilate first.
- Naloxone 400 mcg IV/IM; if no response after 1 min give 800 mcg, repeat 800 mcg, then 2 mg; reassess diagnosis and monitor re-sedation.
- Confirmed cardiac arrest: immediate CPR/standard ALS; naloxone is not routine.

4. MEDICINES - DOSE, VOLUME, TIMING

OXYGEN
- Highest available concentration during CPR.
- After ROSC, titrate to SpO2 94-98%.

ADRENALINE 1:10,000 (100 mcg/mL)
- 1 mg = 10 mL IV/IO.
- PEA/asystole: give ASAP.
- VF/pVT: give after shock 3.
- Repeat every 3-5 min after first dose.

AMIODARONE 30 mg/mL
- After shock 3: 300 mg = 10 mL IV/IO.
- After shock 5: 150 mg = 5 mL IV/IO.

MAGNESIUM SULFATE
- Torsades: 2 g IV over 10 min with pulse; 2 g IV/IO if pulseless.

SODIUM CHLORIDE 0.9%
- 20 mL IV/IO drug flush. Fluid bolus only for a defined indication; reassess.

OTHER INDICATIONS - NOT ROUTINE ARREST DRUGS
- Glucagon: severe hypoglycaemia 1 mg IM; selected beta-blocker toxicity requires expert advice.
- Hydrocortisone: suspected adrenal crisis 100 mg IV/IM.
- Atropine: symptomatic bradycardia 500 mcg IV; repeat every 3-5 min to 3 mg maximum.

5. ROSC: CONFIRM, STABILISE, TRANSFER

- Confirm signs of circulation, then stop CPR and perform ABCDE.
- Maintain airway and waveform capnography; aim normal PaCO2.
- Titrate oxygen to SpO2 94-98%.
- Treat hypotension; target SBP >100 mmHg.
- Obtain 12-lead ECG; identify and treat the cause.
- Control temperature; check glucose and treat seizures.
- Pre-alert and transfer to the appropriate receiving centre; anticipate re-arrest.

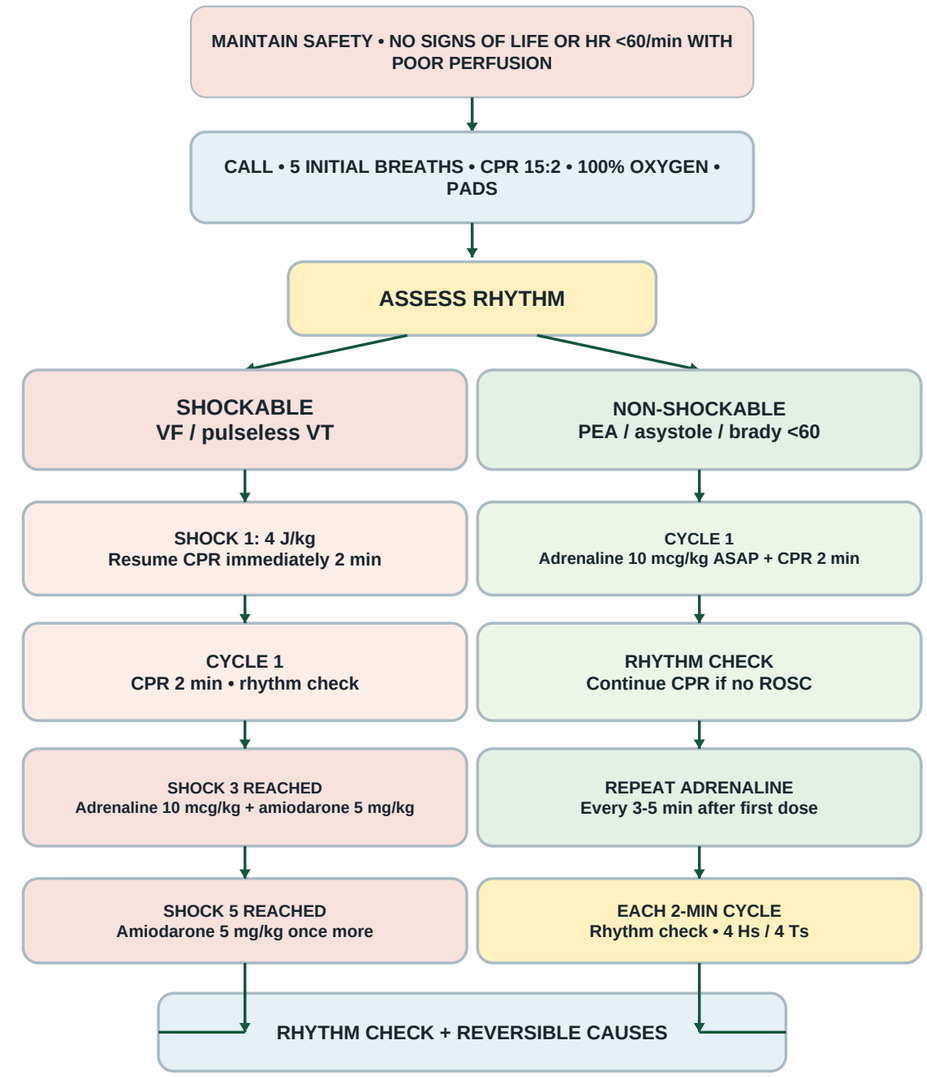
CONSIDER / ESCALATE

- Coronary angiography / PCI when indicated.
- Mechanical CPR for transfer/treatment when appropriate.
- Ultrasound for reversible causes without interrupting CPR.
- Extracorporeal CPR where available and indicated.
- After 3 failed shocks, consider antero-posterior pad position.

PAEDIATRIC ADVANCED LIFE SUPPORT - ONE-PAGE COGNITIVE AID

UK paramedic practice | Core 2-minute cycles, weight-based medicines, reversible causes, special circumstances and ROSC

1. CORE PALS ALGORITHM - CONTINUE 2-MINUTE CYCLES



HOW THE CYCLES + DRUG TIMING WORK

[ ] VF/pVT: shock 4 J/kg, then CPR 2 min. Count shocks, not rhythm checks.
[ ] After shock 3: adrenaline 10 mcg/kg + amiodarone 5 mg/kg during CPR.
[ ] After shock 5: repeat amiodarone 5 mg/kg once. Continue adrenaline every 3-5 min.
[ ] PEA/asystole/brady <60 with poor perfusion: adrenaline 10 mcg/kg ASAP, then every 3-5 min.
[ ] At each rhythm check: shock VF/pVT; do not shock PEA/asystole; if ROSC, start post-ROSC care.
[ ] From shock 5 in refractory VF/pVT: escalate stepwise to 8 J/kg, maximum 360 J.
[ ] If ETCO2 <1.5 kPa, ask why: CPR quality, ventilation, airway/circuit or reversible cause?

2. REVERSIBLE CAUSES - TICK + TREAT

[ ] Hypoxia: open airway; oxygenate and ventilate.
[ ] Hypovolaemia: control loss; 10 mL/kg fluid if indicated, reassess.
[ ] Hyper/hypokalaemia + metabolic: test and treat.
[ ] Hypo/hyperthermia: measure core temperature; correct.
[ ] Thrombosis: coronary or pulmonary pathway.
[ ] Tension pneumothorax: immediate decompression.
[ ] Toxins: antidote/supportive care; seek expert advice.
[ ] Tamponade: trauma/specialist pathway.

3. SPECIAL CIRCUMSTANCES - WHAT CHANGES?

HYPOTHERMIA: KEEP 2-MIN CYCLES
- <30°C: continue CPR/rhythm checks; maximum 3 shocks, then defer further shocks until >30°C.
- <30°C: withhold adrenaline and defer further antiarrhythmic doses.
- 30-35°C: resume shocks; adrenaline every 6-10 min.
- >35°C: standard 3-5 min timing. Prevent heat loss, rewarm and seek specialist/ECPR advice.
- Target normocapnia; avoid hyperventilation.

TORSADES DE POINTES
- Pulse: magnesium sulfate 25-50 mg/kg IV over 10-15 min; max 2 g; may repeat once.
- Pulseless: treat as VF/pVT and defibrillate.

DROWNING / TRAUMA / ANAPHYLAXIS
- Drowning: 5 initial breaths with 100% oxygen; apply hypothermia changes if cold.
- Trauma: treat haemorrhage, hypoxia, bilateral tension pneumothorax and tamponade in parallel.
- Anaphylaxis before arrest: IM adrenaline by age + crystalloid; in arrest use PALS adrenaline.

PREGNANT ADOLESCENT
- Use adult pregnancy modification: manual left uterine displacement if >20 weeks.
- Standard shocks/drugs; call obstetric, paediatric and neonatal support early.

TOXINS: NALOXONE
- Opioid respiratory depression WITH a pulse: ventilate first; use current JRCALC/BNFc age-weight naloxone dose.
- Confirmed arrest: immediate CPR/standard PALS; naloxone is not routine.

4. WEIGHT-BASED MEDICINES + SHOCKS

OXYGEN
- 100% during CPR; after ROSC titrate to SpO2 94-98%.

ADRENALINE 1:10,000 (100 mcg/mL)
- 10 mcg/kg = 0.1 mL/kg IV/IO.
- PEA/asystole/brady <60: give ASAP.
- VF/pVT: give after shock 3.
- Repeat every 3-5 min; maximum single dose 1 mg.

AMIODARONE 30 mg/mL
- 5 mg/kg = 0.167 mL/kg IV/IO after shocks 3 and 5 only.
- Maximum 300 mg after shock 3; 150 mg after shock 5.

FLUID / GLUCOSE
- Balanced isotonic crystalloid or NaCl 0.9%: 10 mL/kg IV/IO if indicated; reassess.
- Known hypoglycaemia: 10% glucose 2 mL/kg IV/IO; recheck 5-10 min.

OTHER INDICATIONS - NOT ROUTINE ARREST DRUGS
- Glucagon severe hypoglycaemia: >=25 kg or >8 y 1 mg IM; <25 kg or <8 y 0.5 mg IM.
- Hydrocortisone adrenal crisis: <1 y 25 mg; 1-5 y 50 mg; >=6 y 100 mg IV/IM.
- Atropine: 20 mcg/kg IV/IO for specified bradycardia indications; max 600 mcg.

Approximate age weights. Amiodarone = 5 mg/kg; glucose 10% = 2 mL/kg (*50 mL cap shown). Verify actual weight/current RCUK + JRCALC charts.

Age	Wt	Adr. mL	Amio mg	Fluid mL	Glucose mL	Glucagon	Hydrocort.	Shock J
<1 mo	3.5 kg	0.35	17.5	35	7	0.5 mg	25 mg	14
1 mo	4 kg	0.4	20	40	8	0.5 mg	25 mg	16
3 mo	5 kg	0.5	25	50	10	0.5 mg	25 mg	20
6 mo	7 kg	0.7	35	70	14	0.5 mg	25 mg	28
1 y	10 kg	1.0	50	100	20	0.5 mg	50 mg	40
2 y	12 kg	1.2	60	120	24	0.5 mg	50 mg	48
4 y	16 kg	1.6	80	160	32	0.5 mg	50 mg	64
6 y	20 kg	2.0	100	200	40	0.5 mg	100 mg	80
8 y	26 kg	2.6	130	260	50*	1 mg	100 mg	104
10 y	30 kg	3.0	150	300	50*	1 mg	100 mg	120
12 y	38 kg	3.8	190	380	50*	1 mg	100 mg	152
14 y+	50 kg	5.0	250	500	50*	1 mg	100 mg	200

5. DURING CPR + ROSC

[ ] CPR 15:2; correct rate, depth and full recoil; minimise pauses; rotate compressor.
[ ] Advanced airway ventilation: infant 25/min; 1-8 y 20/min; 8-12 y 15/min; >12 y 10/min.
[ ] IV/IO access; waveform capnography; check glucose and temperature.

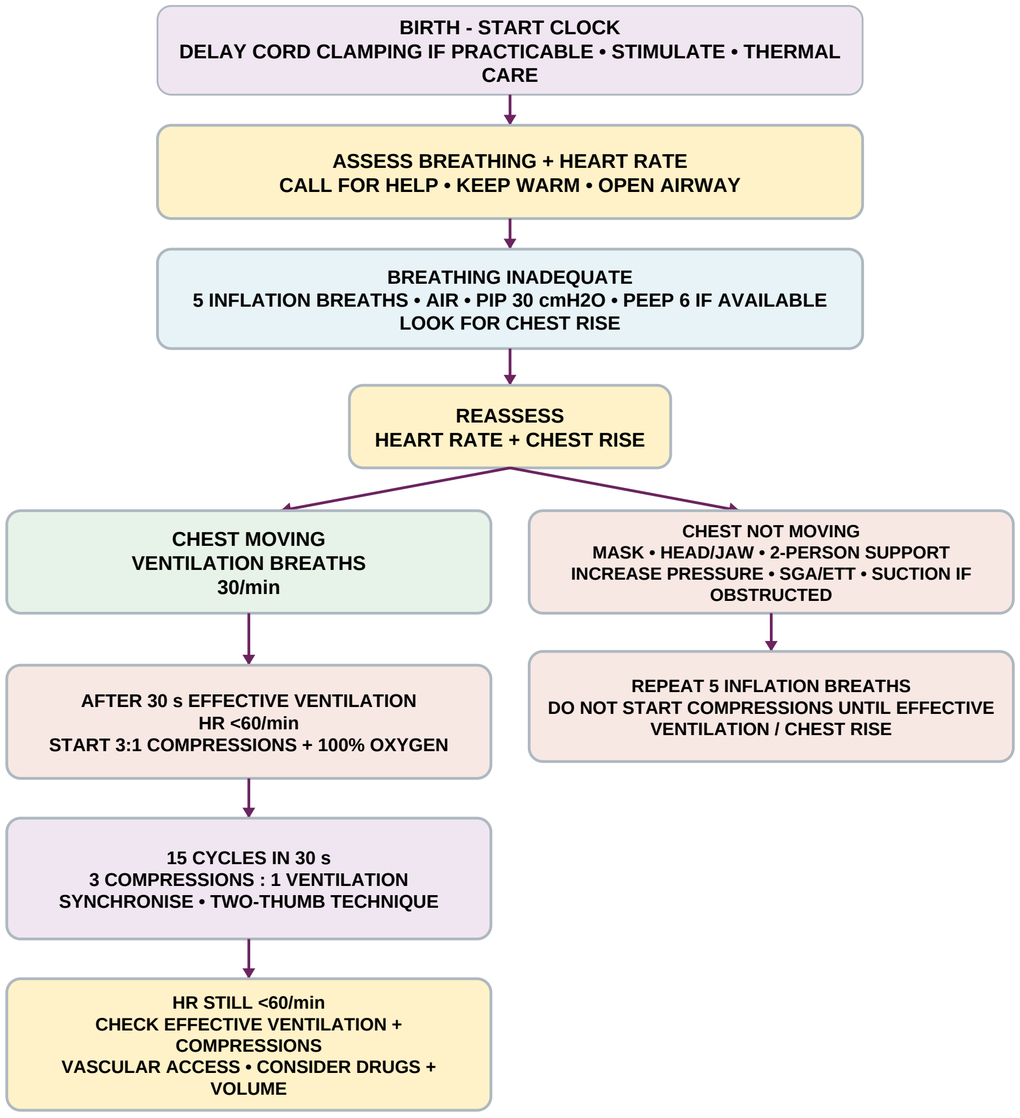
WHEN ROSC IS ACHIEVED

- Confirm circulation, stop CPR, perform ABCDE.
- SpO2 94-98%; normal PaCO2; maintain systolic and mean BP above age 10th percentile.
- Treat cause; glucose control; avoid/manage hypothermia.
- Pre-alert appropriate paediatric receiving team; anticipate re-arrest.
- Provide family support and follow local safeguarding/documentation policy.

NEONATAL ADVANCED LIFE SUPPORT - ONE-PAGE PARAMEDIC COGNITIVE AID

UK pre-hospital practice | Ventilation first | Birth transition, compressions, medicines, thermal care and transfer

1. CORE NEWBORN LIFE SUPPORT ALGORITHM



THE PRIORITY: AERATE THE LUNGS

- Heart-rate improvement is the best sign that ventilation is effective.
- Reassess heart rate and chest rise every 30 seconds. ECG may supplement, not delay, ventilation.
- If unsure whether this is immediate newborn transition or later infant arrest, use the algorithm the team knows while summoning specialist help, then switch clearly and safely.
- Keep interruptions minimal. Continue ventilation during transfer if breathing remains inadequate.

2. FIRST MINUTE + THERMAL CARE

- Term infant: dry, remove wet towels, cover head and use warm environment. Aim 36.5-37.5°C.
- Preterm <32 weeks: without drying, place body in a plastic bag/wrap under radiant heat; keep head accessible.
- If cord intact and resuscitation impractical, clamp within 30 seconds to avoid delaying ventilation. Do not milk the cord if <28 weeks.
- Use pre-ductal SpO2 probe on right hand/wrist. Consider ECG for continuous heart-rate assessment.

3. OXYGEN + PRE-DUCTAL SpO2 TARGETS

- 3 min: 70-75%
- 5 min: 80-85%
- 10 min: 85-95%
- Term/late preterm needing respiratory support: begin with air, then titrate oxygen to target.
- Preterm <32 weeks: use at least 30% oxygen when respiratory support is needed; titrate to target.
- During chest compressions use 100% oxygen; reduce promptly when heart rate recovers and titrate to SpO2.

4. AIRWAY + VENTILATION TROUBLESHOOTING

- Effective ventilation is the intervention most likely to restore heart rate.
- Position the head neutrally; support jaw; use an appropriately sized mask and a two-person technique early.
- If chest does not move: improve seal/position, increase inflation pressure stepwise, consider SGA or tracheal tube, and suction only if obstruction is suspected.
- Consider pneumothorax, congenital airway/lung abnormality or equipment failure if ventilation remains ineffective.

5. ACCESS, DRUGS + VOLUME - VERIFY JRCALC

- Use vascular access only after effective ventilation and compressions have been established and HR remains <60/min.
- Umbilical venous access is preferred where trained/equipped. Use other access according to current JRCALC Plus and local maternity/neonatal policy.
- Adrenaline concentration, weight-based dose, route and repeat interval: verify the current JRCALC Plus Newborn Life Support entry before administration.
- Suspected blood loss/hypovolaemia: consider warmed isotonic crystalloid or blood product and seek neonatal/obstetric clinical support.
- Check blood glucose once immediate resuscitation is controlled; treat using the current neonatal/JRCALC pathway.

6. STABILISATION, TRANSFER + HUMAN FACTORS

- Request neonatal/midwifery critical-care support early and pre-alert the receiving unit.
- Continue airway support, SpO2/ECG monitoring and active thermal care. Avoid hyperoxia and hypocarbia.
- Document time of birth, cord clamping, interventions, pressures/oxygen, heart-rate response, medicines/volume and temperature.
- Keep parents informed where possible. Support parental presence without compromising care.
- Complete team debrief and equipment restock after handover.

7. PARAMEDIC DECISION POINTS

- NEWBORN AT BIRTH: use NLS principles, with ventilation first.
- LATER COLLAPSE / CLEAR POST-TRANSITION INFANT ARREST: use PLS unless a neonatal specialist pathway is explicitly indicated.
- MECONIUM: do not routinely suction. Prioritise ventilation; suction only for suspected airway obstruction.
- PRETERM BREATHING: consider CPAP 5-8 cmH2O where available and within scope.
- ESCALATE EARLY: maternity/neonatal retrieval, clinical support desk and receiving neonatal unit.

Clinical cognitive aid only. Verify the current JRCALC Plus Newborn Life Support guideline, local medicines policy, equipment, scope of practice and neonatal retrieval pathway. RCUK 2025/updated March 2026 framework.